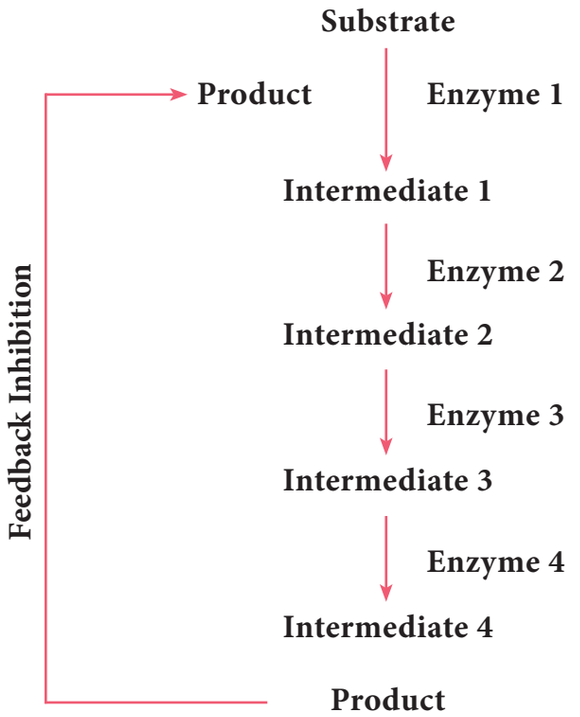
of reactions. This process is used to regulate the synthesis of amino acids (Flowchart 4.2). Example: Threonine deaminase is the first enzyme the conversion Threonine to Isoleucine. Isoleucine inhibits Threonine deaminase through feedback inhibition.



Flowchart 4.2: Feedback Inhibition

4.10.5 Uses of Microbial Enzymes

Microbial enzymes are

- helpful to save energy and prevent pollution
- highly specific
- be immobilized and reused
- inexpensive and more stable
- easily extracted and purified
- genetically manipulated to yield higher quality

Summary

The sum of all chemical reactions within a living organism is known as Metabolism. Biochemical pathway that functions in both anabolism and catabolism are called amphibolic pathways, meaning that they are dual purpose. The energy of catabolic reactions is used to drive anabolic reactions. The energy for chemical reactions is stored in ATP. The chemical reactions are catalyzed by different enzymes. Enzymes catalyze chemical reactions by lowering the activation energy. Most of the cells energy is produced from the oxidation of carbohydrates. During respiration organic molecules are oxidized. Energy is generated from the ETC. In aerobic respiration, O_2 function as the final election acceptor. In anacrobic respiration, the final electron acceptor is an inorganic molecule NO2-, SO42- other than O_2.

Complete oxidation of glucose molecule takes place in 3 sequential reactions.

- Glycolysis occurring in cytoplasm
- Krebs cycle occurring is mito chondrial matrix

ETC (Oxidative Phosphorylation) occurring is inner mitochondrial matrix. In aerobic prokaryotes, 38 ATP molecules can be produced from complete oxidation of a glucose molecule in glycolyins, krebscycle, and ETC. In eucaryotes 36 ATP molecules are produced from complete oxidation of a glucose molecule. In incomplete oxidation of glucose molecules will revolt in fermentation, O2 in anaerobic condition. Various commercial products are produced from pyruvic acid. Lipid can be catabolised by lipase which hydrolyze lipid into glycerol and fatty acid. Then fetly acids are catabolised by Beta oxidation. Proteins can be catabolised by Deamination and

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